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Matter Smart Fan with the Arduino Nano Matter

Learn how to build a Matter fan with speed control using the Arduino Nano Matter

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Overview

This tutorial will teach you how to use the Arduino Nano Matter to create a Matter fan to keep you cool.



Matter fan overview

Thanks to the seamless compatibility of the Nano Matter with almost any Matter network we can easily integrate our fan with Amazon Alexa, Google Assistant, Apple Home, Home Assistant and even any professional custom solution.

We have prepared a short demo in video format in case you are a visual learner.

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Build a Matter-Enabled F...



Hardware and Software Requirements

Hardware Requirements

[Arduino Nano Matter](#) (x1)

Grove motor driver module (x1)

DC motor (x1)

Breadboard (x1)

Jumper wires

Google Nest Hub Max (Thread
Border Router) (x1)

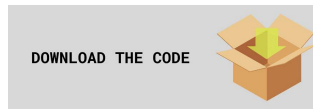
[USB-C® cable](#) (x1)

Software Requirements

[Arduino IDE 2.0+](#) or [Arduino
Cloud Editor](#)

[Google Home App](#)

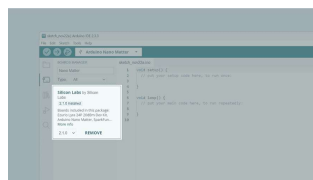
Download the Project Code



Download the complete project code [here](#).

Board Core and Libraries

The **Silicon Labs** core contains the libraries and examples you need to work with the board's components, such as its Matter, Bluetooth® Low Energy, and I/Os. To install the Nano Matter core, navigate to **Tools > Board > Boards Manager** or click the Boards Manager icon in the left tab of the IDE. In the Boards Manager tab, search for `Nano Matter` and install the latest `Silicon Labs` core version.

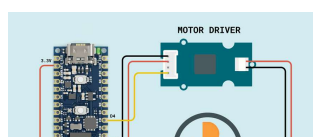


Installing the Silicon Labs core in the Arduino IDE

Project Setup

Schematic Diagram

Use the following connection diagram for the project:

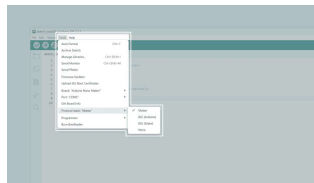


Project wiring diagram

The motor driver module is powered by the **3.3V** Nano Matter output pin. The motor speed is controlled by a PWM signal generated by the pin **D4**.

Programming

In the Arduino IDE upper menu, navigate to **Tools > Protocol stack** and select **Matter**.



Matter Protocol stack
selected

Copy and paste the following sketch:

```

1  #include <Matter.h>
2  #include <MatterFan.h>
3
4  #define fan_pin D4
5
6  MatterFan matter_fan;
7
8  void setup()
9  {
10     Serial.begin(115200);
11     Matter.begin();
12     matter_fan.begin();
13
14     pinMode(BTN_BUILTIN,
15     pinMode(LED_R, OUTPUT)
16     digitalWrite(LED_R, HI
17
18     pinMode(fan_pin, OUTP
19
20     Serial.println("Matte
21
22     if (!Matter.isDeviceC
23         Serial.println("Mat
24         Serial.println("Con
25         Serial.printf("Manu
26         Serial.printf("QR c
27     }
28     while (!Matter.isDevi

```

The structure of this example code is very simple, the main functions are explained below:

In the `setup()` function we initialize the Matter connectivity and the fan output pin.

In the `loop()` function we listen to any fan control command sent from the smartphone or voice assistant and then adjust the fan speed setting the PWM signal accordingly.

Once you uploaded the example code to the Nano Matter, open the Serial Monitor and reset the board.



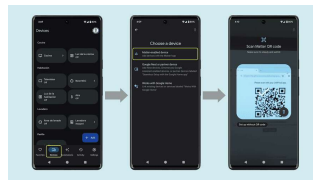
QR Code URL

After the reset you will find on the serial port the URL that generates the QR for the Matter device commissioning.

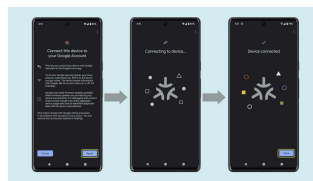
Adding the Device (Commissioning)

Copy and paste the QR code URL on your favorite web browser and a unique QR code will be generated for your board.

Go to your **Google Home** app, navigate to **devices** and tap on **Add**, select the **Matter-enabled device** option and scan the QR code.



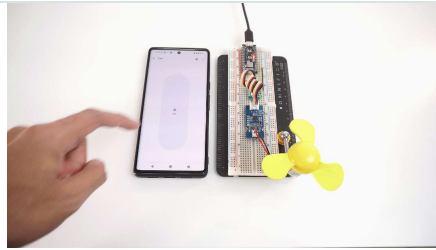
Adding the device to
Google Home app



Your device is successfully
added

Final Results

Finally, you will be able to control the fan from your smartphone, hub or asking your personal assistant.



Conclusion

In this tutorial we have learned how to create a Matter enabled fan that can be controlled from our smartphone and personal assistant. The Nano Matter allows us to seamlessly integrate our own custom or old fan as a commercial product with our current smart home ecosystem.

Next Steps

You can take this solution even further by adding an external knob to adjust the fan speed manually or integrating an AC motor driver to control an actual ceiling fan.

Suggest changes	Need support?	License
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